Raju Krishnamoorthy, Ph.D.

Berlin, Germany | krishnamoorthy@gmail.com | GitHub: notnotraju | GitLab: rkrishnamoorthy | notnotraju.github.io | LinkedIn: rajukrishnamoorthy

Research Engineer

Professional Summary

I am a cryptographic engineer and research mathematician. My recent work has been in cryptographic proving systems, mostly in C++ and Rust, with Python used for research prototypes. I have worked on WASM SIMD optimization, soundness audits, and tests for proving-system code, where small mistakes can compromise the system's security. Before cryptography, I worked in arithmetic geometry, focusing on local systems of geometric origin.

Technical Focus

Languages: C++, Rust, Python.

Performance and systems: WASM SIMD, low-level arithmetic, profiling, benchmarking, cross-platform performance analysis.

Cryptography: Zero-knowledge proof systems, protocol implementation from papers, polynomial commitments, sumcheck, multiscalar multiplication.

Research and correctness: Differential checks, adversarial edge-case generation, code/protocol audit, AI-assisted coding and review, working knowledge of Lean.

Engineering Experience

Aztec 2025.06–present

Applied Cryptography Engineer

- Worked on Barretenberg, Aztec's C++ cryptographic proving library, across low-level finite-field arithmetic, elliptic-curve algorithms, and proof-system code.
- Implemented WASM SIMD field arithmetic, approximately doubling Montgomery-multiplication throughput across tested architectures. Integrated this into Pippenger multiscalar multiplication, improving performance by approximately 30%, and a row-parallel sumcheck path, improving performance by approximately 20%.
- Audited circuit-builder and ECCVM (Elliptic Curve Virtual Machine) logic: compared protocol requirements against implementation behavior, isolated dangerous edge cases and underconstrained logic, and added revealing failure tests and fixes.
- Set up harnesses to attempt to autoformalize the semantic soundness of various circuit pieces.
- Investigated cross-platform performance differences using benchmarks and generated-code inspection to understand compiler scheduling, runtime/JIT behavior, register pressure, and hardware effects.

Irreducible 2024.07–2024.12

Cryptographic Research Engineer

- Implemented research constructions in Python and Rust, producing reference models for Binius-related proof-system components.
- Worked on non-2-primary additive FFTs over binary fields and ring-switching constructions for building small-field polynomial commitment schemes from large-field PCS.
- Worked with the research team to compare protocol variants and improve concrete prover efficiency and proof size.

Academic Research and Teaching

Humboldt Universität Berlin 2022.10–2024.06

Wissenschaftlicher Mitarbeiter Berlin

Bergische Universität Wuppertal 2020.10–2022.10

Wissenschaftlicher Mitarbeiter Wuppertal

University of Georgia 2018.08–2020.08

Limited Term Assistant Professor Athens, GA

Freie Universität Berlin 2016.08–2018.08

NSF Postdoctoral Fellow; supervisor: Hélène Esnault Berlin

- Research focus: local systems of geometric origin and abelian varieties.
- Taught undergraduate and graduate courses in English and German; organized research seminars in English.

Education

Ph.D. in Mathematics, Columbia University 2010–2016

B.S. in Mathematics with Computer Science, MIT 2005–2008

Selected Publications

- R. Krishnamoorthy, J. Yang, and K. Zuo. "A Lefschetz theorem for crystalline representations." To appear in *Algebra & Number Theory*, 2026; arXiv:2003.08906.
- R. Krishnamoorthy, J. Yang, and K. Zuo. "Constructing abelian varieties from rank 2 Galois representations." *Compositio Mathematica* 160(4):709–731, 2024.
- R. Krishnamoorthy and A. Pal. "Rank 2 local systems and abelian varieties II." *Compositio Mathematica* 158(4):868–892, 2022.
- R. Krishnamoorthy. "Correspondences without a core." *Algebra & Number Theory* 12(5):1173–1214, 2018.